

BANG Meeting

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BANG Updates

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Bateman

$$\dot{N}_p = R - \lambda N_p$$

$$\dot{N}_d = \lambda_p N_p - \lambda_d N_d$$

$$N_p(0) = N_d(0) = 0$$

`scipy.integrate.solve_ivp`

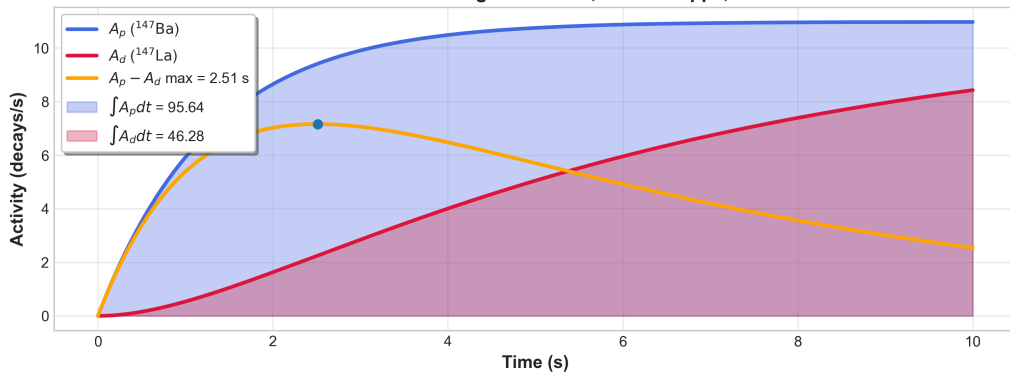
$$^{147}\text{Ba } t_{\frac{1}{2}} = 0.894 \text{ s} \quad ^{147}\text{La } t_{\frac{1}{2}} = 4.06 \text{ s}$$

$$^{148}\text{Ba } t_{\frac{1}{2}} = 0.619 \text{ s} \quad ^{148}\text{La } t_{\frac{1}{2}} = 1.411 \text{ s}$$

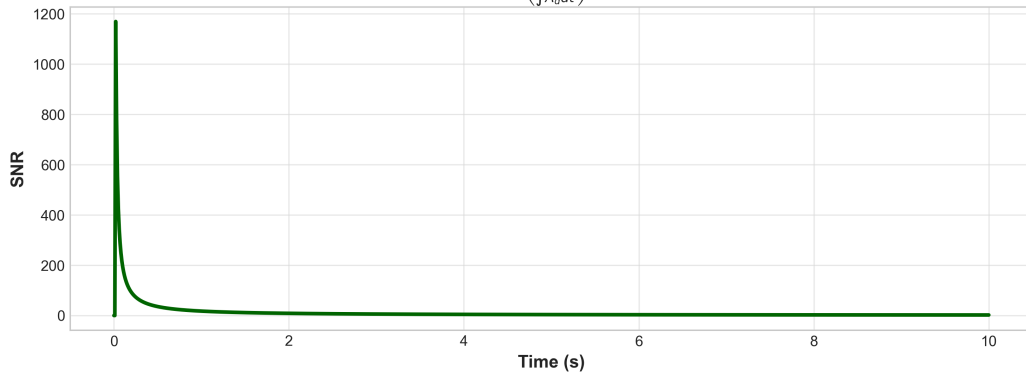
$$\text{maximize SNR}(t_{\text{cycle}}) = \frac{\int_0^{t_{\text{cycle}}} A_p \, dt}{\int_0^{t_{\text{cycle}}} A_d \, dt} \text{ might have to specify minimum contamination}$$

Find maximum net A_p

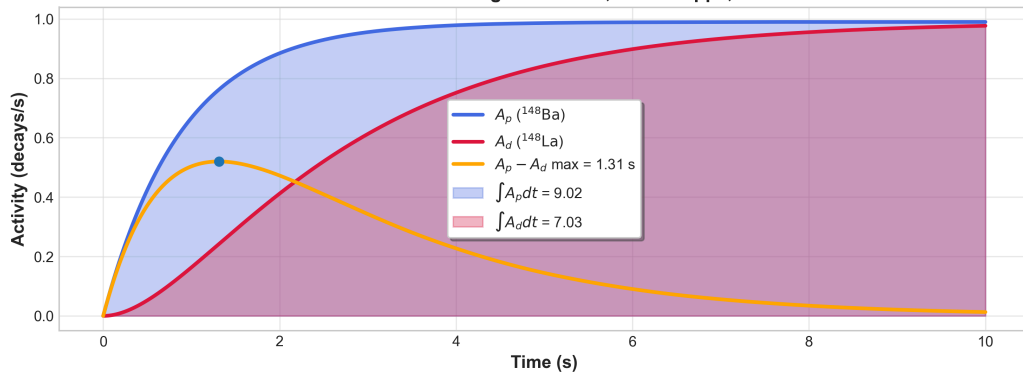
A = 147 Activities with Integrated Areas, R = 10.98 pps, SNR = 2.07



Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle



A = 148 Activities with Integrated Areas, R = 0.99 pps, SNR = 1.28



Signal-to-Noise Ratio $\left(\frac{\int A_p dt}{\int A_d dt} \right)$ Over the Tape Cycle

